

# Virus inactivation by solvent/detergent treatment using Triton X-100 in a high purity factor VIII

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## Abstract

Virus inactivation by solvent/detergent treatment using 0.3% tri-*n*-butyl phosphate and 1% Triton X-100 in the high purity factor VIII concentrate Replenate<sup>®</sup> has been investigated. A wide range of model enveloped viruses were confirmed to be inactivated by >4 to >6 log after 30 min at 22 °C under standard conditions. Using Sindbis as a representative enveloped virus, the effect of various parameters on the inactivation process was tested. Virus inactivation was confirmed to be effective in different batches of product and was not influenced by changing the process conditions with regard to protein and salt concentration or pH. Virus inactivation was effective even at a temperature as low as 4–5 °C. Although solvent/detergent concentration was the most critical parameter, a concentration as low as 0.15% TnBP/0.5% Triton X-100 was still completely effective. At a lower concentration an extended incubation period was required. These studies demonstrate the robustness of this solvent/detergent procedure based on Triton X-100 and allow suitable process limits to be set for this manufacturing step. © 2008 The International Association for Biologicals. Published by Elsevier Ltd. All rights reserved.

**Keywords:** Virus inactivation; Factor VIII; Plasma product; Solvent/detergent; Triton X-100; Process limits

## 1. Introduction

In addition to virus screening, specific virus inactivation or removal steps are incorporated into manufacturing processes for plasma-derived products in order to ensure virus safety [1]. The main steps used include pasteurisation, dry-heat treatment, low pH treatment and virus filtration. The most widely used method is the solvent/detergent (SD) procedure in which a mixture of the organic solvent tri-*n*-butyl phosphate (TnBP) and a non-ionic detergent such as sodium cholate or polysorbate 80 (Tween 80) is used [2–5,13]. This combination of chemicals has the capacity to inactivate such relevant enveloped viruses as human immunodeficiency virus (HIV), hepatitis B virus (HBV), hepatitis C virus (HCV) and West Nile virus (WNV) [4–6].

The manufacturing process for the high purity factor VIII concentrate Replenate<sup>®</sup> uses monoclonal antibody affinity chromatography as a protein purification step and is one of several similar products produced by the ‘Method M’ procedure

[14,16,17,20]. This chromatographic procedure contributes to the safety of the product; however, an SD step, based on Triton X-100, is included to provide a robust virus inactivation step.

While SD treatment is extensively used for virus inactivation, the detergent that is by far the most commonly used is polysorbate 80. This procedure has been used for a wide number of different products. Virus inactivation in plasma products tends to be more rapid when Triton X-100 is used [7]. In the current study, virus inactivation by SD treatment using Triton X-100 has been investigated in a high purity factor VIII product by virus spiking studies with a number of relevant model viruses. This includes testing the effect of varying manufacturing process parameters on virus inactivation in order to test the robustness of this particular version of the SD process.

## 2. Materials and methods

### 2.1. Product

Factor VIII intermediate (cold supernatant) from the Replenate<sup>®</sup> process was obtained from the BPL production

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department. This intermediate is an aqueous extract of cryo-precipitate from which fibrin and fibrinogen had been removed by precipitation under cold conditions. The material was adjusted to 0.8 M NaCl and 50 mM CaCl<sub>2</sub>, and had a pH of 6.5–7.5, conductivity of 55–75 mS and protein concentration ( $A_{280}$ ) of 5–35. For one experiment, intermediate provided by the production department after SD addition and mixing was used. Intermediates from other high purity factor ‘Method M’ products were also tested for comparison.

## 2.2. Virus

Vaccinia (strain Lister, Lister Institute, UK), Sindbis (Lister Institute, UK), herpes simplex virus type 1 (HSV-1; strain HFEM, Leeds University, UK), Semliki Forest virus (SFV; strain 32, Warwick University, UK) and vesicular stomatitis virus (VSV; Indiana serotype, Lister Institute, UK) were propagated in BHK-21 (Clone-13, Leeds University, UK) cells grown in minimal essential medium (MEM) supplemented with 5% newborn calf serum. Infectious cell-culture supernatants were harvested when maximum cytopathic effect had developed and were clarified by centrifugation at  $1000\text{--}2000 \times g$  for 10 min. Infectious cell extracts were prepared by sonication, followed by centrifugation at  $2000 \times g$  for 10 min and filtered at 0.2 or 0.45  $\mu\text{m}$ . In some cases supernatants were concentrated 100-fold using an Easy-Flow ultrafiltration cassette (Sartorius) with a polysulphone membrane of 100,000 molecular weight cut-off. Virus preparations were stored at  $-70^\circ\text{C}$ .

## 2.3. Virus infectivity assays

Virus infectivity was determined by plaque assay on monolayers of BHK-21 cells for Sindbis virus or Vero cells (ICN Biomedicals, High Wycombe, UK) for vaccinia, HSV-1, SFV and VSV using a carboxy methylcellulose overlay. Serial 10-fold dilutions were prepared and aliquots of 0.5 ml assayed on cells in 12-well (3.8 cm<sup>2</sup>) plates. The most concentrated samples assayed were a 1/1000 dilution that had been shown to have no cytotoxicity on the cells and not to interfere with the virus assay. In order to increase the sensitivity of the assay, volumes of 7 ml were assayed on cells in Petri dishes (58 cm<sup>2</sup>). After incubation for 2 days at  $37^\circ\text{C}$ , the plates were stained with methylene blue and the plaques counted. Virus titres in pfu/ml were calculated from sample dilution, assay volume and plaque count. Where virus was undetectable, the titre was calculated assuming one infectious virus unit had been present in the total volume assayed and the final titre expressed as a ‘less than’ value.

## 2.4. Solvent/detergent treatment

Samples (1 ml) of product intermediate were spiked with virus, at a dilution of 1 in 20 or greater, and equilibrated in a water bath at  $22^\circ\text{C}$ . SD was then quickly ( $\sim 1$  s) added from a  $20\times$  concentrated stock solution to give a final concentration of 0.3% TnBP (extraction analysis grade, Merck Ltd)

and 1% Triton X-100 (Ph. Eur, Merck Ltd) and mixed by repeated inversion. Timing was initiated immediately the SD addition had commenced. Samples were taken before the addition of SD and at various time points during incubation at  $22^\circ\text{C}$ . The samples were immediately diluted 1/1000 in PBS in order to stop the inactivation process. In one experiment (Table 5), product intermediate was obtained from the production department after SD addition and inactivation was initiated by the addition of virus. A control was prepared by adding virus to cell-culture medium at a similar dilution. Virus reduction values were calculated by subtracting log virus titres after SD treatment from that in the appropriate product control that had not been treated with SD.

## 2.5. Cytotoxicity and virus interference

The toxicity of the factor VIII intermediate, which contained SD, on the cells used in the virus infectivity assay was tested. A 10-fold dilution series of the intermediate was prepared in PBS and applied to cells grown in 12-well tissue culture dishes as described for the virus plaque assay. The cell monolayer was viewed at the end of the incubation period, after staining. The effect of the factor VIII intermediate on the virus assay was also tested. Virus dilution series were prepared in either a 1/1000 or 1/10,000 dilution of product intermediate in PBS. A control was prepared using PBS alone as diluent. Virus infectivity was then determined by plaque assay as described in Section 2.3.

## 2.6. Stop conditions

The effectiveness of the stop conditions used to terminate virus inactivation by the SD was tested. Virus was incubated in a 1/1000 dilution of factor VIII intermediate, with SD or PBS as a control, for periods of 1, 4 or 24 h. Virus infectivity was then determined by plaque assay as described in Section 2.3.

## 2.7. Effect of process parameters

The effect of altering the conditions for the SD step on virus inactivation was tested. The effect of SD concentration was tested using samples of factor VIII intermediate prepared with different concentrations of SD. Product without SD was also included. Temperature was tested by incubating samples in a water bath at various temperatures over a range of  $4\text{--}30^\circ\text{C}$ . Protein concentration was tested by diluting a special high protein batch of product intermediate to a range of protein concentrations, i.e. 3–40 g/L, using 0.8 M NaCl/0.05 M CaCl<sub>2</sub> as diluent. The protein concentration was determined by absorbance at 280 nm. The effect of pH was tested by adjusting the factor VIII intermediate to pH values of 5.8–7.8 using 0.1 M HCl or 0.1 M NaOH as appropriate. The effect of salt concentration was tested by preparing batches of intermediates with a range of different salt concentrations. Higher salt concentrations were achieved by the addition of extra salt to the standard intermediate and lower

salt concentrations by diluting the standard intermediate with intermediate lacking salt. Salt concentrations were also monitored by conductivity measurement.

### 3. Result

#### 3.1. Control studies

Initial studies were carried out in order to determine a suitable dilution of the product intermediate that had no detrimental effect on the subsequent virus infectivity assay. The factor VIII intermediate, containing SD, was found to be highly toxic to the relevant cells used in the virus assays at a dilution of 1/100 but had no effect at a dilution of 1/1000. The effect of the intermediate at this dilution on the virus assay was then tested. A 1/1000 dilution was confirmed to cause no significant interference, i.e. 0.1 log, to the virus assays.

The effectiveness of the sample dilution step for terminating the virus inactivation process was also tested. It was confirmed that virus was not inactivated to any significant extent (0.2 log) by incubation for short periods of 1 or 4 h, or extended periods of 24 h, in factor VIII intermediate with SD at a 1/1000 dilution. Similar results were obtained when the incubation period was extended to 24 h. Based on these findings it was concluded that diluting the samples to 1/1000 was effective for stopping virus inactivation and that samples could be held for periods of up to 24 h before assay.

#### 3.2. Virus inactivation

SD treatment using Triton X-100 inactivated a range of enveloped viruses in the high purity factor VIII concentrate (Table 1). Virus inactivation was very rapid with at least 4 log of virus inactivation occurring within 1 min for most of the viruses that had been tested in this study. One relevant exception to this was vaccinia virus for which a longer incubation period of at least 10 min was required. HIV has also been shown to be effectively inactivated by this SD combination in other factor VIII concentrates made by the 'Method M' procedure [14]. The effect of the product intermediate alone, i.e. without solvent/detergent, was also tested. No significant virus inactivation, i.e. 0.2 log, occurred when virus was incubated in product intermediate alone, i.e. without SD, for 60 min (Table 4).

Table 1  
Virus inactivation in high purity factor VIII by solvent/detergent treatment using Triton X-100

| Virus    | Envelope | Model characteristics          | Virus inactivation (log) |        |        |
|----------|----------|--------------------------------|--------------------------|--------|--------|
|          |          |                                | 1 min                    | 10 min | 30 min |
| Sindbis  | +        | HCV/WNV                        | 5.4                      | >5.6   | >5.6   |
| SFV      | +        | HCV/WNV                        | 3.9                      | >6.5   | >5.6   |
| HSV-1    | +        | Herpes/large DNA viruses (HBV) | >5.1                     | >5.1   | >5.1   |
| VSV      | +        | Enveloped                      | 6.7                      | >6.7   | >6.7   |
| Vaccinia | +        | Very resistant                 | 2.5                      | 3.8    | 4.4    |

#### 3.3. Effect of manufacturing parameters

Further studies were carried out in order to evaluate this SD procedure in more detail. As the sensitivity to virus inactivation was essentially similar over the range of typical enveloped viruses tested, further studies were conducted using Sindbis virus as a representative enveloped virus and a model for such viruses as HCV and WNV. In all cases the parameters were tested over a range that was significantly greater than that used in production in order to exaggerate any effect on virus inactivation that the parameter might have. By this means the robustness of the virus inactivation process could be assessed.

##### 3.3.1. Product and batch

Virus inactivation by SD treatment in several related factor VIII products, all made by the 'Method M' process, was compared. Virus inactivation was similar in all cases with virus reaching undetectable levels within 2 min (Table 2). This confirms the consistence of virus inactivation in intermediates from different manufacturing facilities under similar conditions.

The inactivation of Sindbis virus in different batches of Replenate<sup>®</sup> was also tested (Table 3). For this experiment, material from a later stage of the manufacturing SD step was used, i.e. after SD addition and mixing for 20 min but prior to the formal incubation step of 30 min. This ensured that the SD addition and mixing step, as performed in production, was also being evaluated in this study. All were from different production batches and thus differed somewhat in their exact composition. However, all were within the manufacturing control limits for this step. Virus inactivation was again very effective and similar in the six batches tested. This further confirms the consistence of this virus inactivation step in routine production. The results also demonstrated that the use of production material in which SD had already been added, rather than adding SD in the laboratory, did not influence the kinetics or degree of virus inactivation.

##### 3.3.2. Solvent/detergent concentration

Of the parameters tested, SD concentration was likely to be critical for effective virus inactivation. The effect of SD concentration on virus inactivation is shown in Table 4.

Table 2  
Inactivation of Sindbis virus by solvent/detergent treatment using Triton X-100: comparison of 'Method M' products

| Product                | Batch | Virus inactivation (log) |       |
|------------------------|-------|--------------------------|-------|
|                        |       | 15 s                     | 2 min |
| Replenate (BPL)        | 1     | 5.0                      | >5.8  |
| 8SM (BPL) <sup>a</sup> | 1     | 5.0                      | >5.9  |
|                        | 2     | >5.4                     | >5.4  |
|                        | 3     | 5.3                      | >5.5  |
| Product-3              | 1     | >5.3                     | >5.9  |
|                        | 2     | 5.4                      | >5.4  |

<sup>a</sup> Manufactured for BPL using cryoprecipitate prepared at BPL.

Table 3  
Inactivation of Sindbis virus in different batches of Replenat by solvent/detergent treatment using Triton X-100

| Batch <sup>a</sup> | Virus inactivation (log) <sup>b</sup> |       |        |
|--------------------|---------------------------------------|-------|--------|
|                    | Time                                  |       |        |
|                    | 15 s                                  | 1 min | 10 min |
| 1                  | 5.3                                   | >5.6  | >5.6   |
| 2                  | >5.6                                  | >5.6  | >5.6   |
| 3                  | 5.6                                   | >5.6  | >5.6   |
| 4                  | 5.6                                   | >5.6  | >5.6   |
| 5                  | >5.6                                  | >5.6  | >5.6   |
| 6                  | >5.6                                  | >5.6  | >5.6   |

<sup>a</sup> Batches of product intermediate were obtained from production after the addition of SD and mixing.

<sup>b</sup> Virus inactivation was determined relative to a medium control spiked with virus (i.e. clearance value).

Inactivation was effective over a range of 0.5–2.0% Triton X-100/0.15–0.60% TnBP. While inactivation was still effective at lower concentrations, i.e. 0.1–0.3% Triton X-100/0.04–0.08% TnBP, an extended incubation period of at least 1 h was needed for satisfactory virus inactivation, i.e. >4 log.

### 3.3.3. Temperature

In view of the fact that temperature can have a major influence on the rate of chemical reactions, this parameter may influence virus inactivation. The effect of temperature on virus inactivation is shown in Table 5. Similar virus inactivation was obtained over a temperature range of 5–30 °C. Virus inactivation was slightly slower at a temperature of 4 °C, nevertheless satisfactory virus inactivation was obtained after 1–10 min.

### 3.3.4. Protein

The effect of protein concentration on virus inactivation was also tested. The New York Blood Centre, who originally developed the SD method, considered lipid concentration to have a possible effect on virus inactivation. However, protein concentration was advocated as a more easily measured surrogate for lipid concentration. In the present study it was shown that virus inactivation was similar over the protein range of 3–40 g/L tested and was similar to that found in the absence of protein (Table 6).

Table 4  
Effect of solvent/detergent concentration on the inactivation of Sindbis virus by solvent/detergent treatment using Triton X-100

| Triton X-100 <sup>a</sup> (%) | TnBP <sup>a</sup> (%) | Virus inactivation (log) |       |        |        |        |
|-------------------------------|-----------------------|--------------------------|-------|--------|--------|--------|
|                               |                       | Time                     |       |        |        |        |
|                               |                       | 15 s                     | 1 min | 10 min | 30 min | 60 min |
| 2.0                           | 0.60                  | 5.2                      | >5.3  | >5.3   | >5.3   | >5.3   |
| 1.0                           | 0.30                  | 5.2                      | >5.3  | >5.3   | >5.3   | >5.3   |
| 0.5                           | 0.15                  | 3.3                      | 3.7   | 4.7    | >5.3   | >5.3   |
| 0.3                           | 0.08                  | 2.3                      | 2.6   | 3.0    | 3.6    | 4.3    |
| 0.1                           | 0.04                  | 1.8                      | 1.9   | 2.3    | 2.6    | 2.9    |
| 0.0                           | 0.00                  | 0.1                      | 0.2   | 0.1    | 0.1    | 0.1    |

<sup>a</sup> Standard conditions are ca. 1.0% Triton X-100 and 0.3% TnBP.

Table 5  
Effect of temperature on the inactivation of Sindbis virus by solvent/detergent treatment using Triton X-100

| Experiment No. | Temperature (°C) <sup>a</sup> | Virus inactivation (log) |       |        |        |        |
|----------------|-------------------------------|--------------------------|-------|--------|--------|--------|
|                |                               | Time                     |       |        |        |        |
|                |                               | 15 s                     | 1 min | 10 min | 30 min | 60 min |
| 1              | 30                            | >5.5                     | >5.5  | >5.5   | >5.5   | >5.5   |
|                | 25                            | >5.5                     | >5.5  | >5.5   | >5.5   | >5.5   |
|                | 22                            | >5.5                     | >5.5  | >5.5   | >5.5   | >5.5   |
| 2              | 19                            | >5.4                     | >5.4  | >5.4   | >5.4   | >5.4   |
|                | 5                             | >5.4                     | >5.4  | >5.4   | >5.4   | >5.4   |
| 3              | 4                             | Nd                       | 3.6   | 4.1    | 4.3    | 4.5    |

Nd, not determined.

<sup>a</sup> Standard conditions are ca. 22 °C.

### 3.3.5. pH

While there was no reason to believe that pH would have any effect on virus inactivation by SD treatment, this parameter was nevertheless tested. The effect of pH over the range of 5.8–7.8 on virus inactivation is shown in Table 7. Virus inactivation was not influenced by this parameter over the range tested.

### 3.3.6. Salt

The effect of salt concentration on virus inactivation was tested (Table 8). This parameter was tested in view of the importance of salt in maintaining the solubility and stability of the factor VIII. Virus inactivation was similar in product without salt addition and after salt addition to 0.6–1.6 M NaCl/0.03–1.0 M CaCl<sub>2</sub>. However, at a salt concentration of 0.4 M NaCl/0.025 M CaCl<sub>2</sub>, virus inactivation was slightly slower, nevertheless 4 to >5 log inactivation was reached by 1–10 min. At this salt concentration, the intermediate was stable over the short periods used for these experiments, however, it clotted when incubated at room temperature over the following 24 h. A lower salt concentration of 0.2 M NaCl/0.0125 M CaCl<sub>2</sub> could not be tested because the product intermediate clotted very rapidly under such conditions.

## 4. Discussion

SD treatment of plasma products is widely used for ensuring the safety, with regard to enveloped viruses, of plasma

Table 6  
Effect of protein concentration on the inactivation of Sindbis virus by solvent/detergent treatment using Triton X-100

| Protein concentration (g/L) <sup>a</sup> | Virus inactivation (log) |       |       |        |        |
|------------------------------------------|--------------------------|-------|-------|--------|--------|
|                                          | Time                     |       |       |        |        |
|                                          | 15 s                     | 1 min | 2 min | 10 min | 30 min |
| 40                                       | >5.8                     | >5.8  | 5.4   | >5.4   | >5.4   |
| 20                                       | >4.9                     | >4.9  | >5.4  | >5.4   | >5.4   |
| 10                                       | >4.7                     | >4.7  | >5.5  | >5.5   | >5.5   |
| 5                                        | >4.9                     | >4.9  | >5.6  | >5.6   | >5.6   |
| 3                                        | >4.8                     | >4.8  | >5.6  | >5.6   | >5.6   |
| 0 <sup>b</sup>                           | >4.8                     | >4.8  | Nd    | Nd     | Nd     |

Nd, not determined.

<sup>a</sup> Standard conditions for the product intermediate are ca. 20 g/L.

<sup>b</sup> Salt solution without any factor VIII intermediate.

Table 7  
Effect of pH on the inactivation of Sindbis virus by solvent/detergent treatment using Triton X-100

| pH <sup>a</sup> | Virus inactivation (log) |       |        |
|-----------------|--------------------------|-------|--------|
|                 | Time                     |       |        |
|                 | 15 s                     | 1 min | 10 min |
| 5.8             | 5.5                      | >5.5  | >5.5   |
| 6.5             | 5.0                      | 5.3   | >5.5   |
| 7.2             | 5.3                      | 4.9   | >5.6   |
| 7.4             | 4.8                      | 5.2   | >5.3   |
| 7.8             | 4.7                      | 4.7   | >5.4   |

<sup>a</sup> Standard conditions are ca. pH 7.2.

products. Many coagulation factors are treated in this way, as well as other plasma-derived products, plasma itself, and products derived from cells in culture. The naturally occurring bile salt sodium cholate was used in the original version of the procedure. Currently, TnBP and the non-ionic detergent polysorbate 80 is the method most commonly used. A version using Triton<sup>®</sup> X-100 is used in the ‘Method M’ process for treating factor VIII, and more commonly for solvent/detergent-treated plasma [10–12]. Recently, an SD procedure using Triton X-100 or Triton X-45 for treating cryoprecipitate has been described [15].

In the current study, virus inactivation by an SD combination based on Triton X-100 has been confirmed to be effective for inactivating a range of model enveloped viruses in the high purity factor VIII Replenate<sup>®</sup>. This is consistent with the results obtained for other products made by the same ‘Method M’ process that all use this same detergent in this virus inactivation step [14,16]. Vaccinia virus was somewhat more resistant than the other viruses tested but was nevertheless effectively inactivated by the full 30 min process treatment period. This virus has previously proved relatively resistant to some virus inactivation procedures such as dry-heat and SD treatment using polysorbate 80. This property is presumably related to the unusual membrane structure that this virus possesses.

Virus inactivation for the conventional enveloped viruses tested was confirmed to be extremely rapid when compared with other variations of the SD procedure. For instance,

Table 8  
Effect of salt concentration on the inactivation of Sindbis virus by solvent/detergent treatment using Triton X-100

| NaCl (M) <sup>a</sup> | CaCl <sub>2</sub> (M) <sup>a</sup> | Conductivity <sup>a</sup><br>(mS cm <sup>−1</sup> ) | Virus inactivation (log) |                 |                 |
|-----------------------|------------------------------------|-----------------------------------------------------|--------------------------|-----------------|-----------------|
|                       |                                    |                                                     | Time                     |                 |                 |
|                       |                                    |                                                     | 15 s                     | 1 min           | 10 min          |
| 1.6                   | 1.0                                | 111                                                 | >5.5                     | >5.5            | >5.5            |
| 0.8                   | 0.05                               | 64                                                  | >5.4                     | >5.4            | >5.4            |
| 0.6                   | 0.03                               | 49                                                  | 5.4                      | >5.4            | >5.4            |
| 0.4                   | 0.025                              | 34                                                  | 2.8                      | 3.7             | >5.4            |
| 0.2                   | 0.0125                             | Nd <sup>b</sup>                                     | Nd <sup>b</sup>          | Nd <sup>b</sup> | Nd <sup>b</sup> |
| 0 <sup>c</sup>        | 0 <sup>c</sup>                     | 5                                                   | >5.4                     | >5.4            | >5.4            |

<sup>a</sup> Standard target conditions are ca. 0.8 NaCl, 0.05 M CaCl<sub>2</sub> and 65 mS cm<sup>−1</sup>.

<sup>b</sup> Sample clotted before the conductivity and virus inactivation could be determined.

<sup>c</sup> Intermediate only, i.e. without salt addition.

when polysorbate 80 is used, the inactivation of >4 logs of a typical enveloped virus can occur in about the first 30 min of the 4–6 h incubation period commonly used for this SD system. However, with Triton X-100, this level of inactivation occurs even more rapidly, e.g. within the first 1–2 min of a 30 min incubation period [7,16]. In both cases an extended incubation period that is well in excess of that required for effective, i.e. >4 log, virus inactivation is used in production. This is to ensure that effective virus inactivation levels are consistently met and to further maximize these levels. In addition the inactivation of some enveloped viruses may be relatively slow, e.g. vaccinia (Table 1) or any ‘unknown’ enveloped virus, and is ensured by extending the incubation period. When Triton X-100 is used for treating plasma directly, to give SD plasma, the concentration of TnBP used is increased from 0.3% to 1.0%. Under these conditions, virus inactivation profiles [10–12] were essentially similar to that seen here with factor VIII in that virus inactivation was rapid and effective over a wide range of viruses.

In the current study, virus inactivation has been further extended to investigate the effect of various process parameters on virus inactivation. While virus inactivation by the Triton X-100 variant of the SD procedure is rapid and effective against a wide range of viruses, it needed to be confirmed that it was consistently effective over the range of parameters that may be encountered in a manufacturing situation. Such an approach is advocated by regulatory authorities for virus inactivation procedures used with plasma products [1,9].

Virus inactivation was confirmed to be consistent for a number of production batches of Replenate and was similar in other products made by the ‘Method M’ process. Of the production parameters tested, the concentration of SD proved to be the most critical. However, virus inactivation was still completely effective even at a lower concentration. This is consistent with the results reported for solvent/detergent plasma where a concentration of 0.4% TnBP and 0.85% Triton X-100 at 30 °C was still effective [10]. In the current study the minimum conditions, determined by testing lower concentrations, that were effective were 0.15% TnBP and 0.5% Triton X-100.

The effect of NaCl and CaCl<sub>2</sub> was also investigated, although in this case the main aim was to test the effect of any instability in the product with regard to clot formation on virus inactivation. However, virus inactivation was effective over the range of salt concentrations tested. However, under certain conditions, the product intermediate was very unstable and clotted rapidly, thus preventing virus inactivation from being tested.

Based on the fact that virus inactivation was relatively resistant to changes in process conditions, the robustness of virus inactivation by the Triton X-100 variant of the SD procedure with plasma products has been demonstrated. The conditions used in production were well within those demonstrated to be effective for virus inactivation in spiking studies, thus ensuring the process was effective in routine use. The method is robust and relatively insensitive to small changes in parameters such as temperature, SD concentration, pH, ionic strength and protein concentration (Table 9). This is similar to other SD



Table 9

Conditions shown to be effective for virus inactivation by solvent/detergent together with manufacturing control limits

| Parameter                           | Virus validated limits | Manufacturing control limits |
|-------------------------------------|------------------------|------------------------------|
| TnBP (% v/v)                        | 0.15–0.60              | >0.25                        |
| Triton X-100 (% v/v)                | 0.5–2.0                | >0.8                         |
| Protein concentration ( $A_{280}$ ) | 3–42                   | 5–35                         |
| pH                                  | 5.8–7.8                | 6.5–7.5                      |
| Conductivity ( $mS\ cm^{-1}$ )      | 34–111                 | 55–75                        |
| Temperature ( $^{\circ}C$ )         | 15–30                  | 18–25                        |
| Time (min)                          | >10                    | >30 <sup>a</sup>             |

<sup>a</sup> Mixing with SD is carried out for 20 min period prior to the formal incubation step.

procedures such as that based on polysorbate 80 [8]. While polysorbate 80 is commonly used in pharmaceuticals as an emulsifier, solubilizer and stabilizer, Triton X-100 is not commonly used in this context. However, neither are considered toxic nor carcinogenic. In addition, both are reduced to near undetectable levels by the manufacturing steps that occur after the SD step. As a detergent for SD virus inactivation, Triton X-100 is being increasingly used for coagulation factors [14,16,17]. Recently, the use of a combination of both 1% polysorbate 80 and 0.3% Triton X-100 with an intravenous immunoglobulin product has been described [18]. However, Triton X-100 may not be suitable for all plasma products. For instance, in the case of SD-treated plasma, the use of Triton X-100 has been reported to result in substantial losses of alpha 2-antiplasmin activity [19].

The relatively wide operating limits that can be set for this SD inactivation step in a manufacturing environment taken together with the fact that virus inactivation with this detergent is very rapid, confirms that this SD combination is suitable for treating biological products. However, the final selection of any particular detergent in the SD procedure will also depend on a number of factors based on the manufacturing process and use of the product.

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